

MERN STACK

DURATION- 4 MONTH

MODULE-1 INTRODUCTION OF JAVASCRIPT ES 5 AND ES6

- Basic JavaScript
- Object Based JavaScript
- Introduction to ES6
- JavaScript Helpers (for Each, filter, map, filter, every, some)
- String Literals
- DE structuring
- Rest parameters & spread operator
- Arrow function
- Default parameter
- Class: Inheritance, constructor
- Promise

MODULE-2 INTRODUCTION TO REACTJS

- What React JS?
- Why use React JS?
- What is Single Page Application?
- Why SPA?
- React JS Version
- REACT DOM
- REACT VIRTUAL DOM
- Set up React JS Environment
- What is CLI: create-react-app project name
- Advantage using CLI
- How to debug React JS application.
- Installation CLI
- Install Node JS
- Install React JS
- Create a hello world example
- Create React JS Project
- Understanding of Folder Structure
- How to debug React JS application
- Introduction of Babel

MODULE-3 BASIC FEATURES OF REACTJS AND COMPONENTS

- React Concepts
- JSX
- Render Elements
- Types of Components
- Class Components
- Life Cycle Method of Class components
- Functional Components
- HOOKS in detail
- Use of HOOK
- What is state
- React Forms
- Components and Props
- State and Lifecycle Handling Events
- Practice on above topics

MODULE-4 ROUTING WITH REACT ROUTER

- Setting Up React router
- Install/ Uninstall React router
- React Router Version 5
- React router Version 6
- Setting up react router
- Understand routing in single page applications
- Working with Browser Router components
- Configuring route with Route component
- Making routes dynamic using Route params
- Working with nested routes
- Navigating to pages using Link and Nav Link Component
- Redirect routes using Redirect Component

MODULE-5 KEY FEATURES OF REACTJS

- Conditional Rendering
- Lists and Keys
- Forms

MODULE-6 EVENT HANDLING IN REACT

- Understanding React event system
- Passing arguments to event handlers

MODULE-7 WORKING WITH FORMS

- Controlled components
- Understand the significance to default Value
- Prop
- Using react ref prop to get access to DOM element

MODULE-8 INTRODUCTION TO REDUX

- what is react redux
- why react redux
- redux install and setup
- actions
- reducers
- store
- high order component
- understanding map state to props and map dispatch to props usage

MODULE-9 REDUX ADVANCED

- Async Actions
- Middleware
- What is redux saga
- Install and set up redux saga
- Working with Saga helpers (Crud Operation)
- Sagas vs Promises

MODULE-10 REACT WITH DESIGN FRAMEWORK

- Material UI for design
- Formik Validation

MODULE-11 REACT API'S

- Introduction https Request
- Introduction Axios NPM
- Introduction Slider NPM
- Integration of any NPM modules
- Introduction of Google Map NPM

NODE JS AND EXPRESS FRAMEWORK

MODULE-1 INTRODUCTION TO NODE.JS

- Introduction to Node.js
- History of Node.js
- Introduction to io.js
- What is Node.js Foundation
- V8 Java Script Engine
- Why Server Side Java Script?

MODULE-2 GETTING STARTED WITH NODE.JS

- Node.js Architecture
- JavaScript Event Loop
- Node.js vs. others Server-Side Frameworks
- Node.js Application Area
- Advantages of Node.js
- Limitations of Node.js

MODULE-3 JAVASCRIPT WITH NODE.JS

- Writing Asynchronous Code
- Blocking vs. Non-Blocking Code

MODULE-4 MODULES

- Understanding Modules
- Built-In Modules
- Creating Module
- Exporting Module
- Importing Modules Node.js CLI and NPM
- Understanding CLI
- Node's Package Manager: NPM
- Local Packages and Global Packages
- Installing, Updating and Removing Module using NPM
- Understanding package .json file

MODULE-5 NODE.JS PACKAGE AND PUBLISHING

- Creating a Node Package
- Publishing Package
- Using published package

MODULE-6 PACKAGE

- Introduction of NPM Package
- Express NPM Package
- Multer NPM Package
- Node Mail NPM Package
- Boot Stripe NPM Package integration
- JSONWEBTOKEN NPM Integration
- Introduction of crypto-JS

MODULE-7 CODE DEBUGGING

- Built-In Debugger
- Node Inspector
- IDE Debuggers

MODULE-8 EXCEPTIONS HANDLING

- Try. Catch
- Call back
- Event Emitters

MODULE-9 EVENTS

- Understanding Events
- Event Emitter class
- Emitting Event
- Listening Event

MODULE-10 STREAMS

- Creating streams
- Streams types

- Readable Stream
- Writable Stream
- Stream Pipe
- Creating Buffer
- Buffer decoding

MODULE-11 FILE SYSTEM AND PATH MODULE

- Introduction to fs Module
- The fs modules operations
- Manipulating Files Create, Open, Read, Write, Close, Rename, Delete
- Manipulating Directives Create, Read, Remove

MODULE-12 EXPRESS FRAMEWORK

- Understanding Express
- Installing Express
- Creating and Running Express App
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MODULE-13 ROUTING AND VIEW ENGINES

- Router Object
- Router Object Methods
- View Engine -Handlebars
- Response Methods

MODULE-14 MIDDLEWARE AND REQUEST PROCESSING

- Middleware
- Express Middleware
- Express Request Processing

MONGODB

MODULE-1

- Understanding NoSQL DB
- NoSQL vs. SQL DB
- Understanding Mongo DB
- Introduction of mongoose Atlas
- MongoDB Data types
- MongoDB Shell Commands
- Understanding db, collection & document
- Understanding Embedded documents
- Querying database Tools & API

- MongoDB Tools
- Introduction to Mongo Chef
- Mongo Chef for database operations

MODULE-2 INDEXING AND RELATIONSHIPS

- Types of Indexes
- Creating an Indexes
- Dropping an Indexes
- Defining Relationships between Documents

MODULE-3 MONGO DB ODM -MONGOOSE AND POSTMAN

- Introduction to Mongoose
- Exploring fundamentals of Mongoose
- Mongoose Models
- Mongoose Data Types
- Mongoose Relationships
- Mongoose CRUD operations

MODULE-4 GITHUB

- Real-time environmental setup with GitHub

REST API WITH MONGOOSE, MONGODB AND POSTMAN

CREATING REST API USING EXPRESS AND MONGOOSE

IMPLEMENTING THE PROJECT DEPLOYMENT OF MEAN APPLICATION